



Lecture Class – I & II & III

Atomic and Molecular Simulations

(CHE F497)

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4th & 6th & 7th Aug 2026

Lecture-I

Handout



1. Course Description:

This course is to equip students with the knowledge of a variety of modern simulation methods at the atomic and molecular levels. The course will deal with several atomic and molecular simulation methods e.g., ab initio methods, molecular dynamics (MD) simulations, Monte Carlo (MC) methods, free energy algorithms, coarse-graining, and multiscale methods. The course will provide hands-on training for simulations using GROMACS or LAMMPS or similar and hands-on simulations using any DFT packages such as VASP, Quantum Espresso or similar. The course also introduces concepts and algorithms based on programming languages (e.g., Matlab, C, C++, Molecular Modelling Software, and Visual Software). Case studies and examples of several real-life applications will be presented for all the simulation techniques.

2. Scope and Objective of the Course:

This course aims to give an overall introduction to various types of particle-based simulation techniques in the sub-micro level phenomena with hands-on training. Currently, with the advances in computational tools, more and more industries/researchers are focusing on using computational tools to study material properties for various applications. The course will include computational multi-scale methods, from first principle ab initio calculations to molecular mechanics, offering potential pathways for studying advanced materials for several applications such as molecular electronics, energy storage, separation, and purification, to name a few. After successful completion of the course, the students will be equipped with the theoretical basis and algorithms of several atomic and molecular simulations techniques (DFT, MD, MC) with their typical application areas and will be able to solve real-world problems by choosing the right simulation techniques.

- **Using computer to simulate the behavior of materials in atomic & molecular scale**
- **Learning the *tools* of modern computational materials science at the atomistic level**
- **Evaluate the tools and their applicability to diverse materials problems**

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3. Course Learning Outcomes:

- Understand and apply fundamental atomic and molecular simulation techniques, including DFT, MD, MC, and multiscale methods.
- Select and apply appropriate simulation approaches to solve real world problems by understanding the phenomena in atomic and molecular scale .
- Analyze and evaluate simulation data to extract meaningful physical and chemical insights for advanced materials applications.

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Course Pre/Co- requisite (if any) & Catalogue / Bulletin Description: No prerequisite (Given in the Bulletin)

4. Text Books:

- **(TB1)** Daan Frenkel, Berend Smit. Understanding Molecular Simulation. From algorithm to Applications. 2002. 2nd edition.
- **(TB2)** David Sholl, Janice A Steckel. Density functional theory: A practical introduction. 2009. Wiley
- **(TB3)** M. P. Allen and D. J. Tildesley, Computer simulation of liquids, 1989, Clarendon Press. NY, USA

5. Reference Books:

1. Marx and Hütter, Ab Initio Molecular Dynamics
2. Maria Velinova, Coarse-Grained Molecular Dynamics, 2017
3. Donald A. McQuarrie. Statistical Mechanics. 2000. University Science Books
4. Statistical Mechanics: Theory and Molecular Simulation Mark Tuckerman, Oxford University Press (2010)
5. Molecular Modelling: Principles and Applications (2nd edition) Andrew R. Leach, Prentice-Hall (2001).

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5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1. Introduction to atomic and Molecular Simulation Methods and recap of Statistical Thermodynamics	L1-L5: Overview of Atomic and Molecular Simulation Methods Fundamental of Statistical Mechanics, Quantum Mechanics, Review of Probability, Micro State, Macro State, Thermodynamic Ensembles	Text Book (TB) TB1-Ch1 TB1-Ch2 TB3-Ch2	• Discuss various aspects of atomic and molecular simulations • Recall the statistical and quantum mechanics used in atomic and molecular simulations
2. Quantum Chemical Methods	L6-L11: Born-Oppenheimer Approximation, Hydrogen and Multielectron Atoms, Hartree-Fock Theory and Basis Sets, Density Functional Theory, Geometry Optimization, Electronic Structure of Atoms and Molecules, Ab Initio MD, Application, A Case Study	TB2-Ch1, Ch3 RB-1	• Discuss the theoretical basis of abinitio calculations • Discuss the significance of basic algorithms and their parameters for a successful simulation
3. Introduction to Quantum Espresso	L12-L14: Hands on simulations using DFT package, e.g., Gaussian, Quantum Espresso	Class notes	• Independently design, run and analyse DFT simulations in a relatively simple scientific problem

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4. Introduction to Molecular Editing and Visualization Software	L15-L17: Making Model Structure, Viewing Simulation Structures and Movies, Making Movies from Simulation Trajectories	Class notes	• Identify a system and create the model structure • visualize the structures and MD/MC trajectories
5. Molecular Dynamics (MD) Simulations	L18-L22: Basics of Molecular Dynamics, Integrators for Newton's Equations, Time step and Timescale Considerations, Thermostat, Barostat, Constant Temperature and Pressure MD, Practical Aspects of Simulations: Simulation of Bulk Phases, Periodic Boundaries, Initialization, Force Computation, Long-Range Interactions, Potential Energy Truncation, Minimum Image Convention	TB1-Ch4 TB3-Ch3	• Discuss the theoretical basis of MD • Discuss the method well enough to be able to deepen the knowledge of various aspects of MD
6. Model and Force Field	L23-L25 : Classical Force Field, Bonded and Nonbonded Interactions, Implicit Solvation, Output and Analysis, Application, A Case Study	TB1-Ch3 TB3-Ch1	• Discuss the significance of model, force field and other parameters for a successful simulation

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7. MD simulation using LAMMPS/GROMACS	L26-L33: Hands on simulations using MD simulation package, e.g., LAMMPS, Gromacs	Reference book 3 + class notes	• Independently design, run and analyse simulations in a relatively simple scientific problem • Learn independent project work and solving open real-world problems
8. Monte Carlo (MC) Methods	L34-L37: Introduction to Monte Carlo Simulation, Markov Chains, The Metropolis Method, Random Number Generator, MC Simulation in Various Ensembles: Canonical Ensemble MC, Isobaric-Isothermal Ensemble, Grand Canonical Ensemble, Rigid Molecules and Orientation Moves, Equilibration, Hybrid MC/MD, Application, A Case Study	TB1-Ch3 TB3-Ch4	• Discuss the theoretical basis of MC • Discuss the method well enough to be able to deepen the knowledge of MC and to run simulations independently

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9. Introducing Coarse Graining and Multiscale Techniques	L37-L40: Kinds of Simplified Models, Advantages of Coarse Graining, Development and Parameterization of Coarse-Grained Models	TB3 + Class notes	• Explain the hurdles of simulations in atomic scale • Describe coarse-grained models • Elaborate the basics of coarse-grain parameterization
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7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Surprised Class Tests (Best 3 out of 4)	20 min	10	In Class	CB
1 Announced Class Test (Pre-MidSem)	25 min	5	In Class TBA	CB
Assignment*-I (Based on Pre-MidSem contents)	1 week	15	TBA	Take Home
Assignments*-II (1 or 2)	2 week	30	TBA	Take Home
Comprehensive Examination	180 min	40	TBA	CB and/or OB

*Assignments: This component involves home assignments throughout the semester based on hands-on training of several simulation techniques.

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8. Chamber Consultation Hour: Will be announced in the class.

9. Notices: All Notices would be put up on Google classroom/Nalanda and/or be announced during the class. All students attending this course **MUST** join google classroom with the class ID TBA. All course related notices and announcements will be announced in the class or be posted in the classroom/Nalanda. Assignments may also be posted in classroom/Nalanda.

10. Make-up Policy: Make-up is granted only for genuine cases in the case of hospitalizations or when complete bed-rest is medically advised with proper justifications, valid documents, and prior permission from the Instructor-in-charge, unless its an emergency. No make-up for class tests and assignments.

11. Note (if any):

*Use of **computer** is mandatory to learn simulations.*

Preliminary requirements

desired



- Basic mechanics and thermodynamics, some quantum mechanics
- Computer is required for exercises
 - computer environment with C or Fortran compiler or MATLAB
- Preferably knowledge on the Unix/Linux environment
- Visualization program
 - VMD (Linux in D208) - <http://www.ks.uiuc.edu/Research/vmd/>
- Molecular Modelling program
 - Avogadro

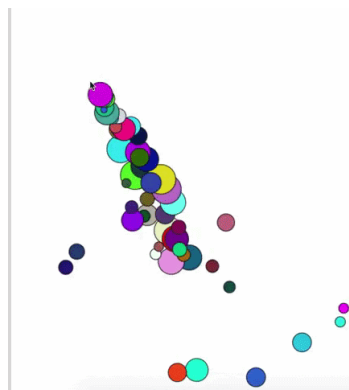
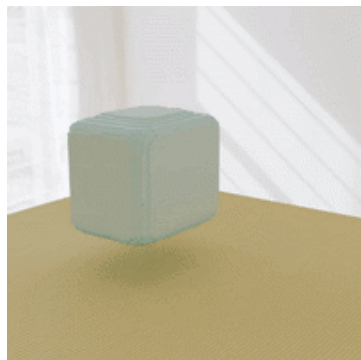


Module-I

**Introduction to
atomic and Molecular Simulation Methods &
Recap of Statistical Thermodynamics**

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Overview of Atomic and Molecular Simulation Methods Fundamental of Statistical Mechanics, Quantum Mechanics, Review of Probability, Micro State, Macro State, Thermodynamic Ensembles

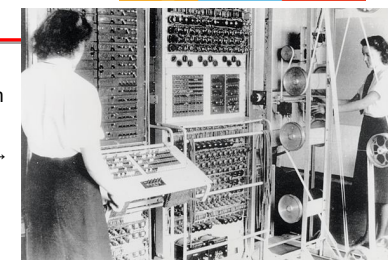
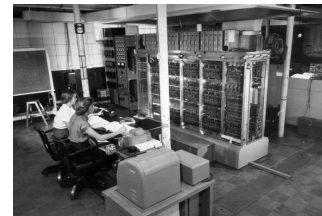


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Introduction

(Pre)History of Computer Simulations

- During the 2nd world War, computing machines had been developed in the development of Nuclear Weapons & code-breaking
- In 1950s, the computers became partly available for nonmilitary use → beginning of Computer simulations
 - N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller (LANL), and E. Teller (Chicago) proposed a general method, suitable for fast computing machines, for investigating properties of liquids
- In 1950's, First simulation of a liquid
 - at Los Alamos National Laboratory, USA Monte Carlo Simulations
- The Los Alamos computer, **MANIAC**, was at that time one of the most powerful



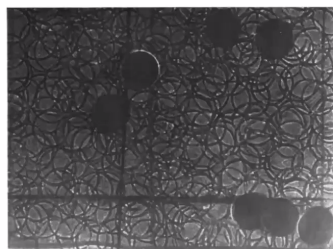
Colossus : the first large-scale electronic computer

Colossus was a set of computers developed by British codebreakers in the years 1943–1945

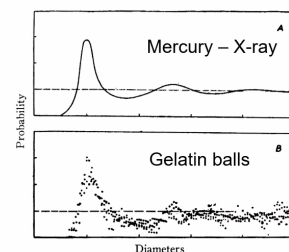
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Introduction: Pre-Computer

- Physical models used to probe statistical properties of liquids



- Gelatin balls suspended in a liquid tank
- Tank shaken to generate typical patterns of separations between balls



- Strong correlation of $g(r)$ between the physical model and mercury

Morrel and Hildebrand, JCP 4, 224 (1936)

A Brief History of Molecular Dynamics Simulations

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Overview

Numerical modeling and computer simulations

- Before, most properties of real materials were predicted on the basis of approximate theories, e.g., →**
 - Van der Waals equation for dense gases**
 - Debye-Huckle theory for electrolytes**
 - Boltzmann equation to describe the transport properties of dilute gases**
- It would be nice if we could obtain essentially exact results for a given model system without having to rely on approximate theories**
- That's what Computer simulations allow us to do**

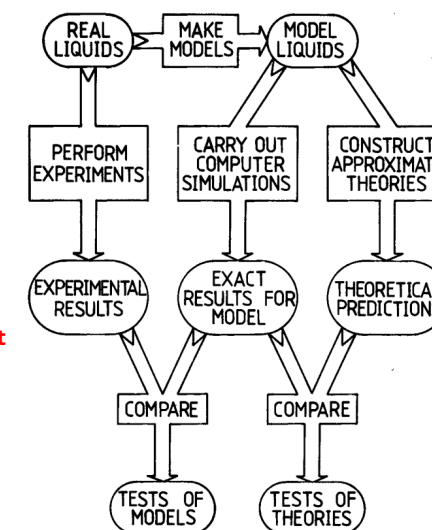


Fig. 1.2 The connection between experiment, theory, and computer simulation.

The First Simulation

Monte Carlo Simulations

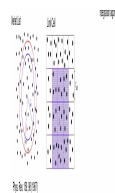
• N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller (LANL), and E. Teller (Chicago)¹ proposed a general method, suitable for fast computing machines, for investigating properties of liquids



1959, 1st MD simulation of a real material (*radiation damage in a crystalline Cu*) by Vineyard at Brookhaven

Short History of Molecular Dynamics Simulation

- **1957, 1959: Alder & Wainwright**, Introduction of basic MD of hard sphere particles
- **1964: Rahman**, Liquid Ar (LJ 6-12, NVE) – First quantitative study with a realistic potential
- **1967: Verlet**, the Verlet numerical integration algorithm & the Verlet neighbor list
- **1974: Stillinger & Rahman**, Water – First study on a realistic system
- **1980, 1981: Andersen, Parrinello-Rahman**, Constant pressure (NPT) MD
- **1984, 1986: Nose, Hoover**, Constant temperature (NVT) MD
- **1985: Car & Parrinello**, ab initio MD (AIMD) based on Density Functional Theory
- **2013: Karplus, Levitt & Warshel**, Nobel Prize in Chemistry



"for the development of multiscale models for complex chemical systems"

The computer – your Virgil in the world of atoms

Newton's classical physics work side-by-side with the fundamentally different quantum physics

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Why Simulations?

Modeling to decide on important outcomes
First Principles Materials Modeling in Industry:

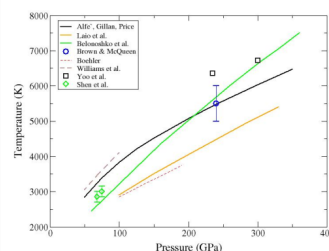
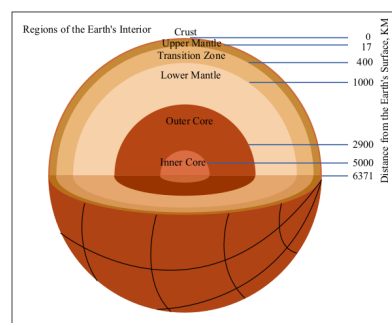
Motorola
 Siemens
 Phillips
 Xerox
 Ford
 Nippon Steel
 Matsushita (Panasonic)
 Ricoh
 TDK
 Hitachi

IBM
 Eastman Kodak
 Electricité de France
 Texas Instruments
 Allied Signal
 Alcoa
 Toyota
 General Motors
 Lucent
 Corning

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Why Simulations?

Calculating things that are difficult to do experimentally: Inside the Earth

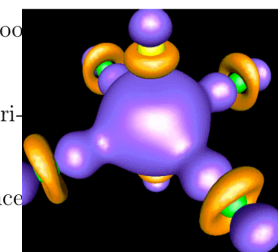


Alfe' et al, *Nature*, 401, 462-464

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Why Simulations?

- Simulation can fill in the gap between theory and the experiment
- Sometimes experiment is too expensive/difficult, sometimes theory too complicated.
- Sometimes simulations are needed to find the proper candidate experiments among many options
- Simulations can also show the important processes among many since they are easy to switch on and off.
- They can also serve as a super microscope when experimental resolution is not good enough.
- However, simulations **alone** are worthless.



Calculating electron densities per orbital

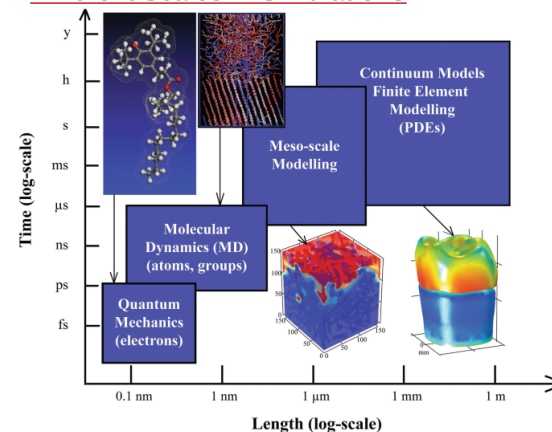
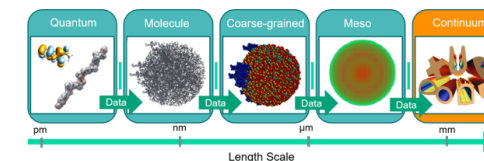
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Atomistic Simulations

- Atomistic simulation models materials at the level of atoms.
- There are different levels of sophistication among these methods, but they share the common feature that
 - Location of atomic nuclei is of principal interest

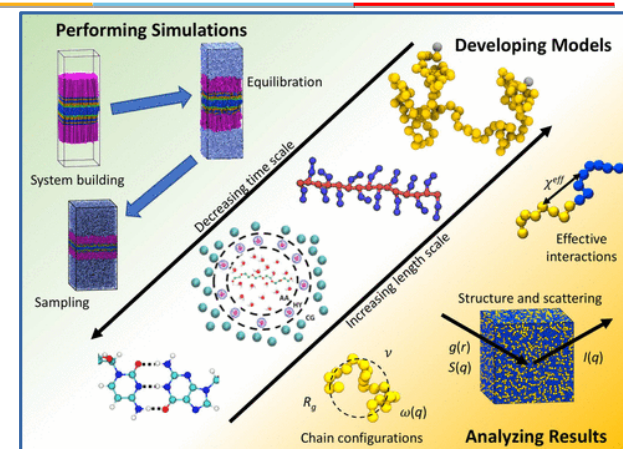
Molecular Simulations

- molecular simulations means that we are interested in methods that deal with molecules.
- atoms are presented explicitly in the simulations
- Materials simulations scales can be divided in 3 classes in terms of simulation timescale
 - microscale, 1ns, atomistic, coarse-grained, kinetic Monte-Carlo
 - mesoscale, 1 μ s, dissipative particle dynamics, etc
 - macroscale, 1s

Different Scales in Simulations**Multi-Scale Simulations**

Molecular and atomistic simulations can be performed in two different philosophies

- Molecular dynamics (MD) or
 - "Real" trajectories
- Monte Carlo (MC)
 - Sampling of configurations



Atomistic & Molecular Simulations

Different levels of methods

1. QM-Based Methods

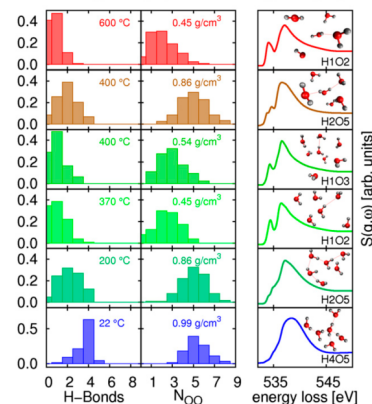
QM methods (DFT, DFTB) can be used for single-point energy evaluation and dynamics of a small system.

- Limited to the range of hundreds of atoms.
- Very short times in dynamical simulations (<100ps) on a supercomputer.
- Most reliable MD method applicable
- To be used when nothing else can (for explicit treatment of chemical reactions)

2. Classical MD

1. Hybrid QM-MM

QM-based methods (ab initio MD)



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Atomistic & Molecular Simulations

Different levels of methods

1. QM-Based Methods

When we have

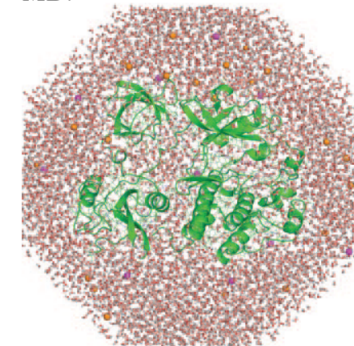
- large systems,
- long time scales or
- long series of events

Most reliable classical Molecular Simulations

Applications: vary from biochemistry to materials sciences. For example bio-molecules and proteins or ion collisions to materials can be studied.

1. Hybrid QM-MM

classical (non-QM) MD.



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Atomistic & Molecular Simulations

Different levels of methods

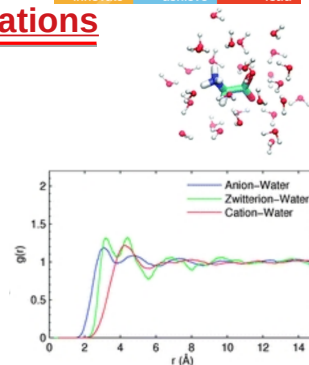
1. QM-Based Methods

2. Classical MD

3. Hybrid QM-MM

QM and classical molecular simulations are applied simultaneously

- Time and size scales can be extended
- Description of relevant properties of different parts of the system
- System divided into parts
 - QM, ab initio MD
 - MM, classical MD
- The parts simulated using different model
- Coupling between the regimes



- QM-MM MD was used to simulate microsolvation structure of Glycine in water.
- In addition core-electron binding energies were evaluated in static DFT-based QM-MM

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Mechanics : Study of position, velocity, force and energy.

Classical Mechanics (Molecular Mechanics)

Molecules (or molecular segments) are treated as rigid object (point, sphere, cube,...)

Newton's law of motion

Quantum Mechanics

Molecules are composed of electrons, nuclei, ...

Schrodinger's equation → Wave function

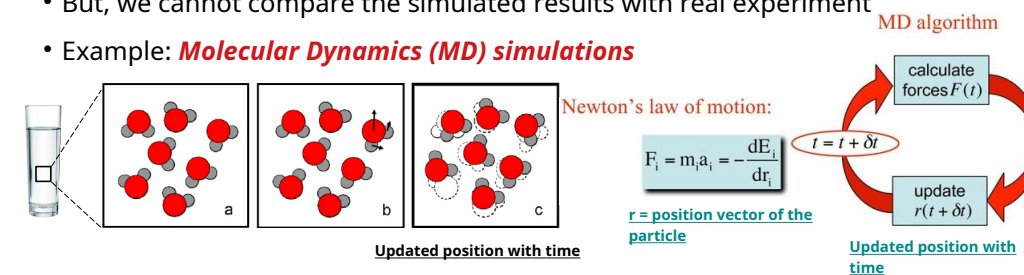
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Thermodynamics and Statistical Mechanics

- Thermodynamics
 - study of the relationships between macroscopic properties
 - Volume, pressure, compressibility, ...
- Statistical Mechanics (Statistical Thermodynamics)
 - how the various macroscopic properties arise as a consequence of the microscopic nature of the system
 - Position and momenta of individual molecules
- Statistical Thermodynamics (or Statistical Mechanics) is a link between microscopic properties and bulk (macroscopic) properties

Why Statistical Thermodynamics?

- We perform computer experiments (MD, MC simulations)
- But, we cannot compare the simulated results with real experiment
- Example: **Molecular Dynamics (MD) simulations**



- No real experiment can provide us with such detailed information
- A typical experiment measures → average properties (average over large number of particles also over time of measurement)

Why Statistical Thermodynamics?

- If we wish to use **computer simulation** as the numerical counterpart of **experiment** → we must know what kind of **averages** we should aim to **compute**
- We need to introduce the **language** of **statistical mechanics**
- Using **statistical mechanics** we can describe the **motion** of **atoms** and **molecules**
- Surprisingly, it turns out to be easier to derive the basic laws of **statistical mechanics** using language of **quantum mechanics**

Statistical Thermodynamics

- **Thermodynamic system:** an assembly of submicroscopic entities, e.g., atoms, molecules, photons, electrons in a very large number of ever changing quantum states
- Each entity has a probability of being in a particular state
- **Probability Function:** the likelihood of that each entity will be in a particular state
- Assembly (System): large collection of identical particles

Statistical Thermodynamics



- Assembly (System): large collection of identical particles
- **Basic postulate:** All possible microstate of an isolated assembly are **equally probable**
- **Microstate:** specified by the number of particles in each energy state
- **Macrostate:** specified by the number of particles in each of the energy levels

Description of States



Macro-states : T, P, V, ... (fewer variables)

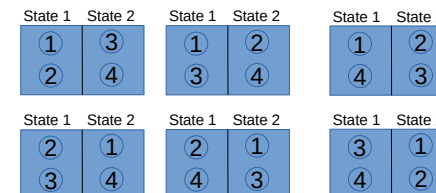
Micro-states : position, momentum of each particles

Each possible outcome is called a "microstate".
The combination of all microstates that give the same number of spots is called a "macrostate".
The macrostate that contains the most microstates is the most probable to occur.

Example: 4 particles system

Macro-state: Having 2 particles in State 1 and 2 particles in state 2. All six have same macro-states but different micro-states

Micro-state: 6 ways in which we can rearrange these 4 particles



MicroStates vs MacroStates

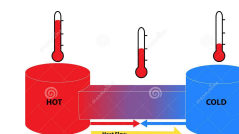
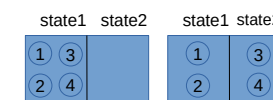


	Microstate	Macrostate
DEFINITION	Microstate is a term that describes the microscopic properties of a thermodynamic system	Macrostate is a term that describes the macroscopic properties of a thermodynamic system
PROPERTIES	Microscopic	Macroscopic
NATURE	Changes of microstate show very slight or no effect on macrostate	Changes in macrostate are the average of large changes of microstates
EXAMPLES	Changes in quantum state	Temperature, pressure, volume and density

Why Statistical thermodynamics?



- Predict the properties of the system - objective
- Because the Large Numbers – Analysis must be done statistically
- Numbers $N \sim 10^{23}$
- Molecules interact via collisions
- Statistical thermodynamics provides a means of averaging
- An assembly tend towards – the most probable state
 - Highest Disorder
 - Highest Entropy
 - Equilibrium



Probability



When toss 4 coins simultaneously, we can get

So, we can get → 4 heads, 2 heads, 1 heads & 0 heads
= 5 macrostates

N = No. Of Coins tossed = 4
 N_1 = No. Of Heads
 N_2 = No. of tails

Macrostate	Individual Microstates	Number of Microstates	P(X)
4 H 0 T	HHHH	1	$\frac{1}{16}$
3 H 1 T	HHHT, HHTH, HTHH, THHH	4	$\frac{4}{16} = \frac{1}{4}$
2 H 2 T	HHTT, HTHT, THHT, THTH, TTHH, HTTH	6	$\frac{6}{16} = \frac{3}{8}$
1 H 3 T	HTTT, THTT, TTHT, TTTH	4	$\frac{4}{16} = \frac{1}{4}$
0 H 4 T	TTTT	1	$\frac{1}{16}$

Fundamental methodology of statistical mechanics



- **A thermodynamic system:** An assembly of subatomic entities (atoms, molecules, photons, electrons etc.) in a very large number of changing states.
- Each entity has a probability of being in a particular state.

Fundamental methodology of statistical mechanics



- A probability function describes the likely hood of each entity in a particular state.
- All micro-states of an isolated assembly of identical particles are equally probable.

total number of degenerate states of the system is $\Omega_1(E_1) \times \Omega_2(E_2)$

Degeneracy of a subsystem 2 is $\Omega_2(E-E_1)$

measure of the degeneracy of the subsystems that is additive.

take the (natural) logarithm of the degeneracy. $\ln$

$$\ln \Omega(E_1, E - E_1) = \ln \Omega_1(E_1) + \ln \Omega_2(E - E_1)$$

subsystems 1 and 2 can exchange energy & Every energy state of the total system is equally likely

that maximizes $\ln \Omega(E_1, E - E_1)$. The condition for this maximum is that

$$\left(\frac{\partial \ln \Omega(E_1, E - E_1)}{\partial E_1} \right)_{N, V, E} = 0$$

We have a system 1 which is in thermal equilibrium with a large heat bath (2), $E = E_1 + E_2$

Weekly interacting system: can exchange energy but sum of energies of two subsystems is total energy

or, in other words,

$$\left(\frac{\partial \ln \Omega_1(E_1)}{\partial E_1} \right)_{N_1, V_1} = \left(\frac{\partial \ln \Omega_2(E_2)}{\partial E_2} \right)_{N_2, V_2}$$

We introduce the shorthand notation

$$\beta(E, V, N) \equiv \left(\frac{\partial \ln \Omega(E, V, N)}{\partial E} \right)_{N, V}$$

$$\beta(E_1, V_1, N_1) = \beta(E_2, V_2, N_2)$$

Clearly, if initially we put all energy in system 1 (say), there will be energy transfer from system 1 to system 2

until equation $\beta(E_1, V_1, N_1) = \beta(E_2, V_2, N_2)$ is satisfied

equilibrium is reached, $\ln \Omega$ of the total system is at a maximum

After all, the second law of thermodynamics states that the entropy of a system N , V , and E is at its maximum when the system is in thermal equilibrium. There are many ways in which the relation between $\ln \Omega$ and entropy can be established. Here we take the simplest route

$$S(N, V, E) \equiv k_B \ln \Omega(N, V, E)$$

$\ln \Omega$ is somehow related to the thermodynamic entropy S

we take the simplest route; we simply

define the entropy to be equal to $\ln \Omega$. In fact, for (unfortunate) historical reasons, entropy is not simply equal to $\ln \Omega$; rather we have

Lecture-III

Statistical Mechanics

- we have formulated **statistical mechanics** in purely **quantum mechanical terms**
- **Entropy (S)** is related to density of states of a system with energy E , volume V and number of particles N
 - $S(N, V, E) \approx k_B \ln \Omega(N, V, E)$

The degeneracy of the energy levels, i.e., the number of eigen states with energy E , particles N and system volume V
- Helmholtz free energy is related to the partition factor Q (i.e., sum over all quantum states of the Boltzmann factor)
- Every quantum state of a many particle system with energy E is equally likely to be occupied
- Such an average over all possible quantum states of a system is called an ensemble average

Thermal Equilibrium between subsystems



Clearly, if initially we put all energy in system 1 (say), there will be energy transfer from system 1 to system 2

until equation $\beta(E_1, V_1, N_1) = \beta(E_2, V_2, N_2)$ is satisfied

$$\beta(E, V, N) \equiv \left(\frac{\partial \ln \Omega(E, V, N)}{\partial E} \right)_{N, V}$$

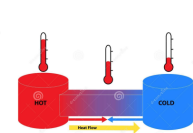
$$S(N, V, E) \approx k_B \ln \Omega(N, V, E)$$

$$= \delta S / \delta E k_B$$

$$= \delta S / (\delta S / T) k_B$$

$$= 1 / (k_B T)$$

temperature is



$$1/T = \left(\frac{\partial S}{\partial E} \right)_{V, N}$$

If we use the same definition here, we find that

$$\beta = 1 / (k_B T)$$

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Statistical Mechanics in Quantum Mechanical Terms



- Entropy is related to the density of states of a system with energy E, volume V and number of particles N

$$S(N, V, E) \approx k_B \ln \Omega(N, V, E)$$

$$\ln \Omega = S(N, V, E) / k_B$$

$$= - (1/T) E / k_B$$

$$\Omega = \exp (- E / k_B T)$$

$$\frac{1}{T} \equiv \left(\frac{\partial S}{\partial E} \right)_{N, V}$$

Change in entropy
Number of particles N and volume V constant
Change in internal energy

$$\text{or, } \partial S = \frac{1}{T} \partial E$$

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in one specific quantum state i with energy E_i



Degeneracy of a subsystem 2 is $\Omega_2(E-E_i)$

subsystems 1 and 2 can exchange energy

We have a system 1 which is in thermal equilibrium with a large heat bath (2), $E = E_1 + E_2$
Weekly interacting system: can exchange energy but sum of energies of two subsystems is total energy

Probability to find a system (A) in a particular state (i)

$$P_i = \frac{\Omega_2(E-E_i)}{\sum_j \Omega_2(E-E_j)}$$

Specific quantum state i with Energy E_i

$$E = E_1 + E_2$$

Average energy of the system at a given temperature T

$$\langle E \rangle = \sum E_i P_i$$

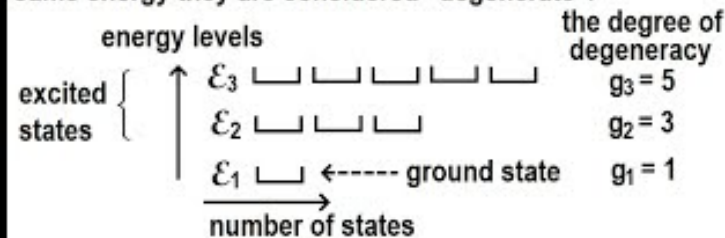
$$P_i = \frac{\exp(-E_i/k_B T)}{\sum_j \exp(-E_j/k_B T)}$$

This is the well-known Boltzmann distribution

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What is a Degenerate Quantum State

When particles are confined to a "container" they will experience quantum states (various ways in which they can be arranged). When different quantum states have the same energy they are considered "degenerate".



- The degeneracy of the energy levels, i.e., the number of eigen states with energy E, particles N and system volume V

$$\Omega(E, V, N)$$

- A system with fixed N, V, E is **equally likely** to be found in any of its eigen states Ω

(E)

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Statistical Mechanics in Quantum Mechanical Terms



• Entropy

$$S(N, V, E) \approx k_B \ln \Omega(N, V, E)$$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{V, N}$$

$$\text{or, } \partial S = \frac{1}{T} \partial E$$

$$\ln \Omega = S(N, V, E) / k_B$$

$$= - (1/T) E / k_B$$

$$\Omega = \exp(-E / k_B T)$$

$$F = -k_B T \ln Q$$

• Helmholtz free energy (F) is related to partition function

• Partition function (Q) is sum over all quantum states i of the boltzmann factor $\exp(-E_i / k_B T)$

$$F = -k_B T \ln \left(\sum_i \exp(-E_i / k_B T) \right)$$

workhorse of equilibrium statistical mechanics

Statistical Mechanics (recap)



- we have formulated **statistical mechanics** in purely **quantum mechanical terms**
- Entropy (S)** is related to density of states of a system with energy E, volume V and number of particles N

$$S(N, V, E) \approx k_B \ln \Omega(N, V, E)$$

- Such an average over all possible quantum states of a system is called an ensemble average

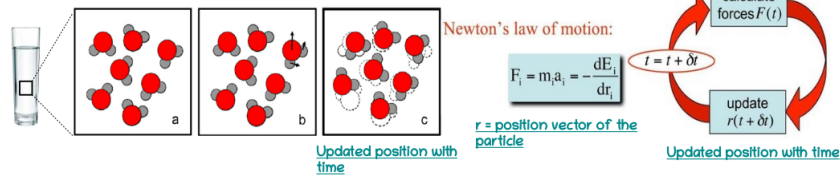
- In MD simulations, we measure time average with assumptions:

- The time sufficiently long
- The time average does not depend on initial conditions

Can we arrive at the macroscopic properties of the system?



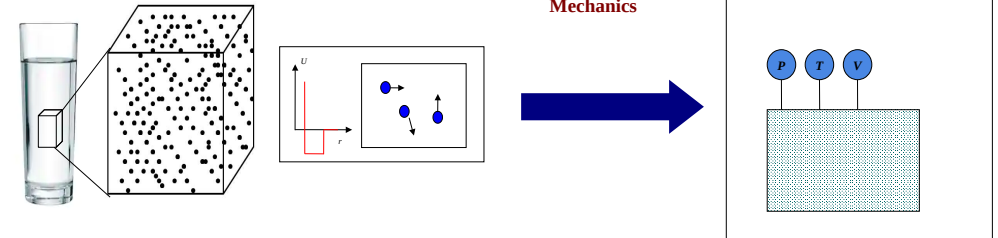
- We perform computer experiments (MD, MC simulations)
- But, we cannot compare the simulated results with real experiment
- Example: **Molecular Dynamics (MD) simulations**



- No real experiment can provide us with such detailed information
- A typical experiment measures → average properties (average over large number of particles also over time of measurement)

Many different microstates all correspond to the same macrostate

Can we arrive at the macroscopic properties of the system?



- we can calculate the macroscopic behaviour of the system by averaging over the corresponding microstates.
- We can derive thermodynamics from the quantum behaviour of atoms and molecules

If the value of some quantity X in the i th microstate is X_i , and the probability that the system is in that microstate is p_i ,

X in the macrostate is the **ensemble average** $\langle X \rangle = \sum_i p_i X_i$

all allowed microstates are equally likely.

(Allowed or accessible means having the same volume, particle number and total energy as the macrostate.)

Degeneracy of a subsystem B is

$$\Omega_B(E-E_i)$$

Probability to find a system (A) in a particular state (i)

$$P_i = \frac{\Omega_B(E-E_i)}{\sum_j \Omega_B(E-E_j)}$$

We have a system A which is in thermal equilibrium with a large heat bath (B),

$$E = E_B + E_A$$

Weekly interacting system: can exchange energy but sum of energies of two subsystems is total energy

Specific quantum state i with Energy E_i

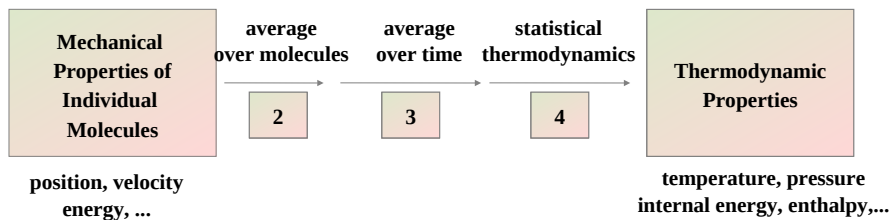
$$E = E_B + E_A$$

Average energy of the system at a given temperature T

$$\langle E \rangle = \sum_i E_i P_i$$

Equilibrium Macroscopic Properties

- Properties are consequence of average of individual molecules
- Properties are invariant with time $\rightarrow$ Time average



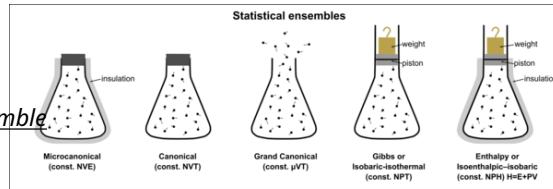
Ensemble

- Every quantum states of a many body system with energy E is equally likely to be occupied
- Such an average over all possible quantum states of a system is called an ensemble average

In statistical mechanics, an ensemble (also statistical ensemble) is an idealization consisting of a large number of virtual copies (sometimes infinitely many) of a system, considered all at once, each of which represents a possible state that the real system might be in.

In other words, a statistical ensemble is a probability distribution for the state of the system.

- 1) Microcanonical or constant - NVE
- 2) Canonical or constant - NVT
- 3) grand canonical or constant - μVT ensemble
- 4) isothermal-isobaric or constant - NPT
- 5) Isoenthalpic-isobaric - NPH



THANK YOU